

## **Title**

Sales and Distribution Analysis of Nassau Candy Distributor Using Data Analytics

## **Abstract**

This paper analyzes the Nassau Candy Distributor dataset containing over 10,000 sales transactions to evaluate sales, profitability, regional performance, products, and shipping efficiency using Python, Pandas, Streamlit, and visualization techniques.

## **Introduction**

The project transforms transactional sales data into actionable business insights through exploratory data analysis and an interactive dashboard.

## **Problem Statement**

Organizations need efficient reporting to identify profitable products, regional demand, shipping performance, and operational bottlenecks.

## **Objectives**

Analyze sales, profit, products, regions, customer behavior, and shipping KPIs while supporting data-driven decisions.

## **Dataset**

The dataset contains 10,194 records and 22 columns including order details, product information, sales, cost, gross profit, lead time, shipment status, region, and division.

## **Methodology**

Data collection, cleaning, exploratory data analysis, KPI calculation, visualization, dashboard development, and business insight generation.

## **Technologies**

Python, Pandas, NumPy, Streamlit, Plotly, Jupyter Notebook.

## **Results**

The dashboard helps identify top-performing products, profitable regions, shipping delays, and business trends for improved operational efficiency.

## **Conclusion**

The project demonstrates how business intelligence dashboards enable informed decision-making by converting historical sales data into meaningful insights.